Year 11 Higher

# REVISION PACK

## Algebra 9

**Name:** _____

**Class:** _____

## Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)

**Worked example: finding the formula, then using it backwards**

$m$ is directly proportional to the cube of $r$. When $r = 2$, $m = 40$

(a) Find a formula for $m$ in terms of $r$. (b) Work out the value of $r$ when $m = 135$

| | | | |
|---|---|---|---|
| (a) | Step 1 | Write with $k$ | $m = kr^3$ |
| | Step 2 | Put in $r = 2$, $m = 40$ | $40 = k \times 8$ |
| | Step 3 | Find $k$ | $k = 5$ |
| | Step 4 | Answer | $m = 5r^3$ |
| (b) | Step 1 | Put in $m = 135$ | $135 = 5r^3$ |
| | Step 2 | Divide by 5 | $r^3 = 27$ |
| | Step 3 | Cube root | $r = 3$ |
| | Step 4 | Answer | $r = 3$ |

### Practice: finding the formula, then using it backwards

**1.1** $m$ is directly proportional to the cube of $r$. When $r = 2$, $m = 24$

(a) Find a formula for $m$ in terms of $r$. (b) Work out the value of $r$ when $m = 81$

**1.2** $m$ is directly proportional to the cube of $r$. When $r = 3$, $m = 54$

(a) Find a formula for $m$ in terms of $r$. (b) Work out the value of $r$ when $m = 128$

**1.3** $m$ is directly proportional to the square of $r$. When $r = 4$, $m = 80$

(a) Find a formula for $m$ in terms of $r$. (b) Work out the value of $r$ when $m = 125$

**1.4** $m$ is directly proportional to the cube of $r$. When $r = 2$, $m = 16$

(a) Find a formula for $m$ in terms of $r$. (b) Work out the value of $r$ when $m = 250$

**1.5** $m$ is directly proportional to the cube of $r$. When $r = 4$, $m = 320$

(a) Find a formula for $m$ in terms of $r$. (b) Work out the value of $r$ when $m = 135$

**1.6** $m$ is directly proportional to the square of $r$. When $r = 3$, $m = 36$

(a) Find a formula for $m$ in terms of $r$. (b) Work out the value of $r$ when $m = 100$

## Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)

**Worked example: inverse proportion to x²**

$y$ is inversely proportional to $x^2$. When $x = 2$, $y = 9$. Work out the value of $y$ when $x = 3$

| | | |
|---|---|---|
| **Step 1** | Write with $k$ | $y = \frac{k}{x^2}$ |
| **Step 2** | Put in $x = 2$, $y = 9$ | $9 = \frac{k}{4}$ |
| **Step 3** | Find $k$ | $k = 36$ |
| **Step 4** | Put in $x = 3$ | $y = \frac{36}{9}$ |
| **Step 5** | Answer | $y = 4$ |

### Practice: inverse proportion to x²

**1.7** $y$ is inversely proportional to $x^2$. When $x = 3$, $y = 8$ Work out the value of $y$ when $x = 6$

**1.8** $y$ is inversely proportional to $x^2$. When $x = 2$, $y = 20$ Work out the value of $y$ when $x = 4$

**1.9** $y$ is inversely proportional to $x^2$. When $x = 5$, $y = 4$ Work out the value of $y$ when $x = 2$

**1.10** $y$ is inversely proportional to $x^2$. When $x = 4$, $y = 5$ Work out the value of $y$ when $x = 2$

**1.11** $y$ is inversely proportional to $x^2$. When $x = 1$, $y = 36$ Work out the value of $y$ when $x = 3$

**1.12** $y$ is inversely proportional to $x^2$. When $x = 10$, $y = 1.5$ Work out the value of $y$ when $x = 5$

## Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)

**Worked example: direct proportion to √x**

$y$ is directly proportional to $\sqrt{x}$. When $x = 16$, $y = 12$. Work out the value of $y$ when $x = 25$

| | | |
|---|---|---|
| **Step 1** | Write with $k$ | $y = k\sqrt{x}$ |
| **Step 2** | Put in $x = 16$, $y = 12$ | $12 = k \times 4$ |
| **Step 3** | Find $k$ | $k = 3$ |
| **Step 4** | Put in $x = 25$ | $y = 3 \times 5$ |
| **Step 5** | Answer | $y = 15$ |

### Practice: direct proportion to √x

**1.13** $y$ is directly proportional to $\sqrt{x}$. When $x = 4$, $y = 10$ Work out the value of $y$ when $x = 36$

**1.14** $y$ is directly proportional to $\sqrt{x}$. When $x = 25$, $y = 15$ Work out the value of $y$ when $x = 64$

**1.15** $y$ is directly proportional to $\sqrt{x}$. When $x = 9$, $y = 6$ Work out the value of $y$ when $x = 81$

**1.16** $y$ is directly proportional to $\sqrt{x}$. When $x = 16$, $y = 8$ Work out the value of $y$ when $x = 100$

**1.17** $y$ is directly proportional to $\sqrt{x}$. When $x = 4$, $y = 14$ Work out the value of $y$ when $x = 9$

**1.18** $y$ is directly proportional to $\sqrt{x}$. When $x = 36$, $y = 30$ Work out the value of $y$ when $x = 4$

## Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)

**Worked example: what happens to y?**

$y$ is inversely proportional to $x^2$. $x$ is doubled. Write down what happens to $y$.

| | | |
|---|---|---|
| **Step 1** | $x$ doubled, so $x^2$ | $2^2 = 4$ |
| **Step 2** | Inverse: do the opposite | $\div 4$ |
| **Step 3** | Answer | $y$ is divided by $4$ |

### Practice: what happens to y?

**1.19** $y$ is directly proportional to $x^2$. $x$ is multiplied by 3. Write down what happens to $y$.

**1.20** $y$ is inversely proportional to $x^2$. $x$ is multiplied by 2. Write down what happens to $y$.

**1.21** $y$ is directly proportional to $x^3$. $x$ is multiplied by 2. Write down what happens to $y$.

**1.22** $y$ is inversely proportional to $x$. $x$ is multiplied by 5. Write down what happens to $y$.

**1.23** $y$ is directly proportional to $\sqrt{x}$. $x$ is multiplied by 4. Write down what happens to $y$.

**1.24** $y$ is inversely proportional to $x^2$. $x$ is multiplied by 10. Write down what happens to $y$.